

Influenza — English boarding school, 1978

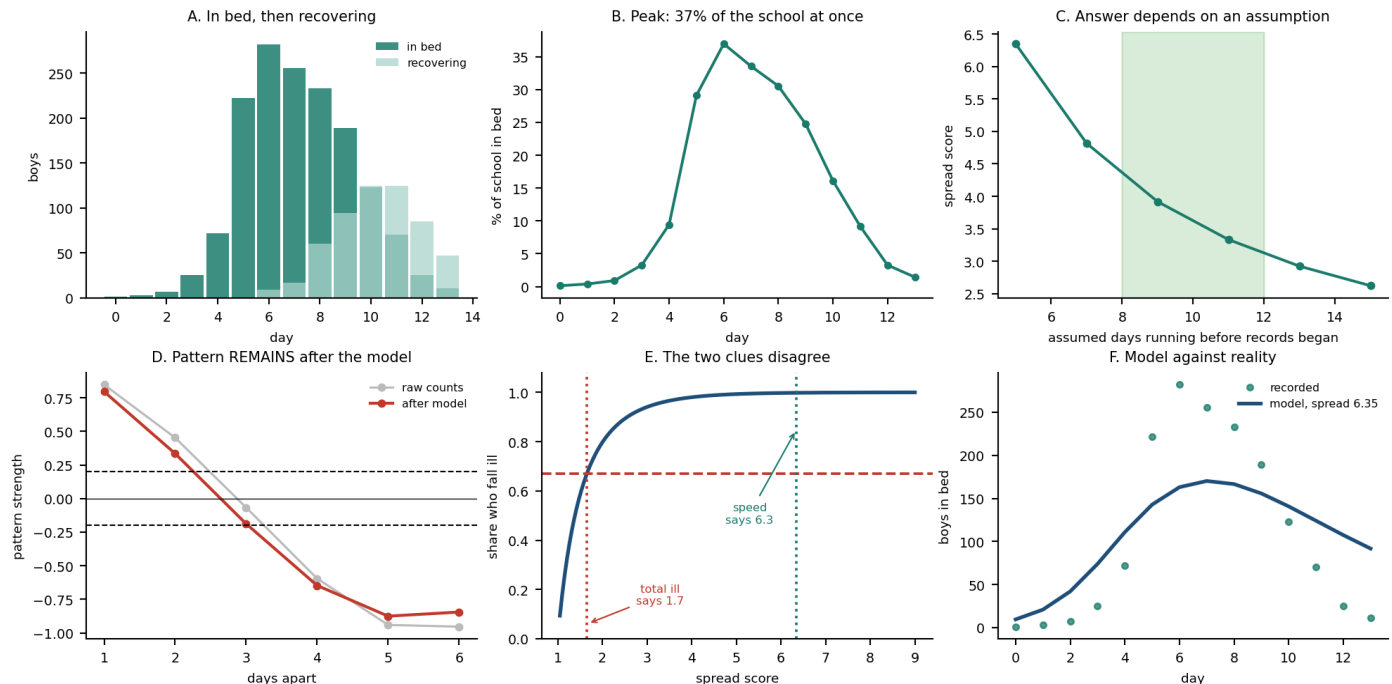
Dossier 3 of 4 · CAVEATED ARM · BMJ 1978;1(6112):587; Avilov et al. (2024) J R Soc Interface 21:20240394

READ THIS FIRST — this outbreak is included as a test of whether our checks can catch a bad case. They did. Its contagiousness figure should not be quoted beside the other three as an equal.

Findings, decisions, and where each one stands

Finding	Value in RAW data	Concern	Decision taken	Status in CANONICAL
Records start mid-outbreak	index boy ill 15–18 Jan, records begin 22 Jan	model would assume it started on day one	assume 11 days already running (inside the documented window)	HANDLED — but the answer is sensitive to it (panel C)
'In bed' is not 'infectious'	boys infectious ~2 days, in bed ~5 days	treating them as the same overstates the outbreak length	v6: model a separate confined-to-bed group	RESOLVED — structure corrected
Counts jump enormously	variance/mean 104	far larger than the other outbreaks	reflects the sharp arc, not noise; forgiving count used	RESOLVED — understood and handled
Pattern remains after the model	lag-1 +0.796	model is genuinely missing something	a change-over-time allowance would need 14 numbers for 14 days	NOT RESOLVED — allowance would swamp the data
Two clues disagree	total ill says 1.66; speed says 6.35	no single contagiousness value satisfies both	none available — published attempts hit the same wall	NOT RESOLVED — disclosed instead
Very few records	14 daily tallies	any answer will be imprecise	none — this is simply what exists	NOT RESOLVABLE — smallest of the four

This shows that two concerns were fixed and three cannot be. The three that cannot are properties of the surviving records, not mistakes — which is why this outbreak is labelled rather than discarded.



This shows that compare panel D here with panel D in the Ebola and measles dossiers. Theirs are green and flat; this one is red and tall. That single difference is the entire reason this outbreak is treated separately.

Why no single answer exists

Way of judging contagiousness	Answer	Source
Count how many eventually fell ill (512 of 763)	1.66	Kermack-McKendrick relation, 1927
Watch how fast it spread through the school	6.35	this analysis
Published model built specifically for this dataset	8.14	Avilov et al. (2024) J R Soc Interface
Typical influenza across all settings	1 – 4	Ahmad et al. (2025) Sci Rep, flagging 8.14 as implausibly high

This shows that four reasonable approaches give four different answers spanning a factor of five. That is not a modelling failure on our part; a 2024 paper exists purely to document that no model fits this outbreak properly.

VERDICT — KEEP, LABELLED. Structure corrected where possible; three limits remain that no method can remove. Retained because it demonstrates the diagnostics detect a genuinely hard case rather than glossing over it. Run the model once without it to confirm it is not distorting the other three.